

Will Humans Look Physically Different in the Future?

A Pathway-Attribution Proposal for Population-Level Physical Phenotypes

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Source-bound Survey–Idea–ExperimentDesign research proposal
Design-only scientific study; no observed results are reported

Abstract—Claims about how future humans will look are often presented as if visible change had one cause: evolution. That framing is scientifically inadequate. A difference between population cohorts can reflect transgenerational allele-frequency change, developmental plasticity, technology or treatment, demographic composition, or a change in measurement practice. These pathways have different time scales, causal meanings, data requirements, and ethical risks. This paper converts the broad question into a falsifiable research proposal: can a pre-registered pathway-aware model provide a better calibrated and more transportable account of selected population-level physical phenotypes than an age, birth-cohort, site, and measurement-process baseline? The proposed Phenotype Pathway Attribution Atlas is a noninterventional longitudinal study using only eligible, consented records. It analyzes pre-specified phenotype families separately, including congenital third-molar agenesis as a bounded dental case study, de-identified craniofacial scalar measurements, and harmonized anthropometry. The protocol prevents a common error in the wisdom-tooth narrative by separating congenital agenesis from eruption, impaction, extraction, and treatment. Optional inherited-feature summaries are subject to separate consent and stringent population-structure sensitivity analyses; they are never used to rank individuals or human groups. Temporal and external-site holdouts, calibration tests, measurement-era ablations, negative controls, and governance release gates define what would count as support. The proposal does not forecast a universal future appearance, does not claim that any trend has been observed, and does not make clinical or reproductive recommendations. Its contribution is an auditable way to replace visually compelling but unsupported speculation with conditional, population-level scientific inference.

Index Terms—human phenotype, developmental plasticity, population genetics, third-molar agenesis, cohort effects, prediction calibration, demographic composition, research governance

I. INTRODUCTION

It is reasonable to expect that human populations in the future will not be identical to those observed today. It is not reasonable to convert that general possibility into a drawing of a universal future face, body, or dentition. Physical phenotype is the result of development within an environment, inherited variation, disease and treatment history, material and social conditions, and the way bodies are sampled and measured. A

cohort difference can therefore be real while its popular explanation is wrong. This distinction is especially important when a claim implicitly treats geographic, ancestry, or demographic categories as fixed biological types.

The term evolution is frequently used too broadly. In a population genetic sense, evolution requires a change in inherited variant frequencies across generations. In contrast, developmental plasticity denotes phenotypic change during ontogeny in response to biological and environmental conditions. Technology may alter phenotype through dentistry, orthodontics, medicine, nutrition, assistive devices, or surgery without making that phenotypic change inherited. Migration and changing cohort composition can alter a population average with no within-lineage biological transformation. Finally, a new scanner, clinical referral rule, or measurement protocol can make a trend appear even when the underlying distribution is unchanged.

The prompt's reference to wisdom teeth is a useful illustration of the problem. Third-molar agenesis is a congenital developmental outcome, while wisdom-tooth eruption, impaction, extraction, and orthodontic management are different outcomes. Carter *et al.* synthesized 92 studies comprising 63,314 subjects and estimated a heterogeneous worldwide agenesis rate of 22.63 percent, with estimates ranging from 5.32 to 56.0 percent [2]. This is evidence that agenesis is a measurable and heterogeneous phenotype. It is not evidence that every population is currently evolving toward tooth loss, that a smaller jaw causes congenital agenesis, or that the frequency must rise in the future.

The same caution applies to genetic forecasts. Modern genomic data make it possible to examine associations with complex traits, but association does not remove population structure, environmental correlation, or transferability problems. Berg *et al.* reported that previously prominent polygenic-adaptation signals for height were strongly attenuated or absent when reexamined with independent UK Biobank estimates, and documented confounding by population stratification [4]. Accordingly, polygenic-score contrasts are not a safe substitute for causal or predictive evidence about visible human differ-

ences.

This paper presents a design-only proposal rather than a claim of completed research. It asks a deliberately bounded question: *after outcome definitions and analysis decisions have been locked, does a longitudinal model that represents demographic composition, developmental exposure, technology or treatment, measurement process, and optional inherited features separately improve calibration and transportability over a cohort-only baseline?* This is a question about population-level research inference. It does not predict an individual appearance and it does not authorize a conclusion about a group merely because the group appears in a dataset.

II. BACKGROUND, SURVEY, AND RESEARCH GAP

A. Pathways that must not be merged

Human developmental systems can respond to conditions during ontogeny. Ponzi *et al.* review hormonal mediation of developmental plasticity and emphasize that changes in phenotype can arise during development through endocrine systems responsive to environmental conditions [1]. That evidence supports a modeling distinction: a population trend in an anthropometric or craniofacial measure may be associated with developmental exposures without representing an inherited change in allele frequency. The source does not specify a universal direction for any future human feature.

At longer time scales, human cultural practices can affect environments in which genetic evolution occurs. Laland *et al.* describe gene–culture coevolution as an important framework for connecting cultural activities, selection environments, and human genetic change [3]. This permits technology, diet, and other cultural variables to be considered as candidate environmental pathways. It does not justify treating a present-day technology association as proof of selection or as a forecast of an inherited trait.

Population composition is a separate pathway. If age distribution, parental birthplace distribution, migration pattern, recruitment source, or survival changes over time, a cohort mean can change even when no individual process has changed. This requires explicit covariates, site effects, and temporal validation. The 1000 Genomes Project reference is a reminder that genotype data require careful quality control and documented structure rather than superficial biological labels [5].

The final pathway is measurement and intervention. Orthodontic treatment can affect dental alignment; extraction removes a tooth after development; medical care and nutrition can affect developmental conditions; and a new radiographic or imaging protocol can alter ascertainment. These variables may be important predictors, but a predictive association is not a causal effect estimate. Without a separately valid causal design, this study will describe associations and forecast performance only.

B. Bounded dental endpoint and genetic caution

The third molars are the most frequently affected teeth in congenital agenesis, but this fact should increase, not reduce,

measurement discipline. The meta-analysis by Carter *et al.* shows considerable geographic and demographic heterogeneity [2]. A record of no visible third molar is not automatically a record of agenesis. The tooth may not yet have erupted, may be impacted, may have been extracted, or may have an unrecoverable history. These alternatives carry different biological and clinical meanings.

The proposed dental endpoint therefore requires an age-adequate developmental assessment and a provenance field that explicitly records congenital absence, known extraction, impaction, eruption status, and treatment history. If a source cannot support that separation, it cannot contribute to the confirmatory dental analysis. This conservative choice can reduce sample size, but it prevents a visually memorable narrative from outweighing endpoint validity.

The proposed study does not deny genetic contribution to physical phenotype. It refuses to assume that such contribution can be read directly from population labels or from a score optimized in another sample. Berg *et al.* provide a direct warning: previous height-adaptation results were susceptible to residual structure and should be interpreted cautiously [4]. A genotype summary may therefore enter the design only after separate consent, documented quality control, relatedness-aware procedures, and analysis that tests whether the finding survives reasonable choices about structure. If it does not, the block is removed. A simpler model with transparent limits is preferable to an impressive model that encodes unresolved confounding.

C. Frozen research gaps

The Survey freezes three linked gaps. **GAP-FP-01** is pathway conflation: future-phenotype claims often merge genetic change, plasticity, technology, composition, and measurement. **GAP-FP-02** is outcome conflation in dental narratives: congenital agenesis is commonly discussed beside extraction and jaw-related outcomes without a common endpoint definition. **GAP-FP-03** is validation weakness: cohort-based forecasts often omit temporal transport, external-site transport, calibration, and sensitivity to population structure or measurement change. The project is designed to address these gaps, not to answer every question about human evolution.

III. RESEARCH QUESTIONS AND PLANNED CONTRIBUTIONS

The primary research question is: *Does a preregistered pathway-aware longitudinal model improve out-of-sample calibration and proper scoring for one or more selected population-level physical phenotypes beyond an age, birth-cohort, site, and measurement-process baseline?*

The supporting questions are whether developmental variables improve prediction after the baseline is locked, whether treatment records resolve endpoint misclassification, whether gains persist across later periods and independent sites, and whether any inherited-feature block remains stable after structure and relatedness sensitivity analyses. The planned contributions are a pathway taxonomy, a strict dental outcome

TABLE I
PATHWAYS THAT CAN PRODUCE APPARENT PHYSICAL DIFFERENCES AND THE INFERENCE EACH PATHWAY PERMITS

Pathway	Example observable inputs	Permitted inference in this study	Prohibited shortcut
Inherited population change	Separately consented, quality-controlled genotype summaries; family and structure information	A candidate predictive block may be evaluated after ancestry-aware sensitivity analyses.	Treating a polygenic-score contrast as a ranking of groups or a deterministic appearance forecast.
Developmental plasticity	Time-aligned developmental exposures and health or nutrition indicators where consented	A developmental context can be associated with a phenotype and may improve prediction.	Calling an ontogenetic response inherited evolution.
Technology and treatment	Dental treatment and extraction history, measurement-appropriate care access proxies	Treatment history can separate congenital from post-developmental outcomes.	Calling an observational treatment coefficient causal.
Demographic composition	Eligibility, recruitment, site, calendar period, and composition descriptors	A cohort average can be decomposed and audited for compositional change.	Treating a changing sample as a changing biological type.
Measurement process	Instrument, protocol, assessor, referral pathway, and measurement era	Method changes can be modeled or can invalidate an endpoint.	Calling a protocol artifact a biological trend.

definition, a common baseline, locked transport tests, and a communication boundary against individual appearance prediction and population ranking.

IV. IDEA SOURCE CHECKPOINTS AND DIRECTION SELECTION AUDIT

The Idea stage compared three directions. A Secular Phenotype Trend Catalogue would inventory cohort differences in teeth, faces, and anthropometry. It is useful descriptively but cannot attribute a difference to a distinct pathway. It was therefore not selected. The selected *Phenotype Pathway Attribution Atlas* uses longitudinal records, separate predictor blocks, and falsification gates. It addresses all three frozen gaps. A universal future- human visual generator was rejected because it has no observable universal target and would produce deterministic, stereotype-prone claims.

No Monte Carlo tree search, autonomous evolutionary optimization, or simulated multi-agent debate was executed. The selection record is transparent rather than fictional. The selected idea fails if pathway-aware models do not improve locked holdout performance, if improvements disappear after endpoint harmonization or sensitivity analysis, or if governance gates cannot be met.

V. PROBLEM DEFINITION, ASSUMPTIONS, AND HYPOTHESES

A. Target population and endpoints

Let Y_{ijk} denote phenotype k for person i at visit or age-interval j . Each endpoint has its own eligibility rule and is analyzed separately. The first confirmatory endpoint is congenital third-molar agenesis, denoted by A_i . It is defined only when an age-adequate source record supports a developmental determination:

$$A_i = \mathbb{I}\{\text{congenital third-molar absence is documented}\}. \quad (1)$$

The indicator in (1) is not assigned from an absent tooth in a later clinical record. Known extraction, impaction, eruption status, and treatment are separate variables. This does not prevent descriptive analyses of those outcomes; it prevents them from entering the same outcome label.

The second family comprises de-identified craniofacial scalar measures, such as pre-registered linear distances or ratios, available only from approved existing data. The third family includes standardized anthropometric measures where an instrument and protocol can be harmonized or modeled. Raw facial images, free-text appearance labels, and subjective attractiveness ratings are excluded. The design has no requirement that all phenotype families be available at every site; the prespecified cohort eligibility table will identify which sites support which endpoint.

B. Model classes

The baseline model represents age or developmental stage, birth cohort, site, and measurement process. It intentionally includes variables that can generate a spurious-looking trend:

$$\eta_{ijk}^{(0)} = \alpha_k + g_k(t_{ij}) + \gamma_{k,c(j)} + u_{k,s(i)} + \delta_{k,m(ij)}. \quad (2)$$

In (2), $\eta_{ijk}^{(0)}$ is the baseline linear predictor for phenotype k , $g_k(\cdot)$ is a preregistered flexible function of age or developmental stage t_{ij} , $\gamma_{k,c(j)}$ is the birth-cohort effect, $u_{k,s(i)}$ is the site effect, and $\delta_{k,m(ij)}$ represents the measurement process. The chosen link function is binary or continuous as appropriate to the endpoint.

The pathway-aware model augments, but does not replace, that baseline:

$$\eta_{ijk}^{(1)} = \eta_{ijk}^{(0)} + \mathbf{x}_{ij}^\top \boldsymbol{\beta}_k + \mathbf{z}_{ij}^\top \boldsymbol{\theta}_k + \mathbf{q}_{ij}^\top \boldsymbol{\phi}_k + \mathbf{r}_i^\top \boldsymbol{\psi}_k. \quad (3)$$

Here, $\mathbf{x}_{ij}$ is the demographic-composition block, $\mathbf{z}_{ij}$ the developmentally time-aligned exposure block, $\mathbf{q}_{ij}$ the technology or treatment block, and $\mathbf{r}_i$ an optional inherited-feature summary. The associated coefficient vectors are $\boldsymbol{\beta}_k$, $\boldsymbol{\theta}_k$, $\boldsymbol{\phi}_k$, and $\boldsymbol{\psi}_k$. Equation (3) is a predictive, multivariable representation. It is not a causal structural equation; in particular, $\boldsymbol{\phi}_k$ is not a treatment-effect estimator.

The quantity evaluated at a future or held-out observation is a conditional predictive distribution:

$$\hat{p}_{ijk} = \Pr(Y_{ijk} \mid \mathcal{H}_{ij}, \mathcal{M}_k^{(1)}), \quad (4)$$

where $\mathcal{H}_{ij}$ is the record history allowed by the locked prediction horizon and $\mathcal{M}_k^{(1)}$ is the pathway-aware model for endpoint k . All imputation, transformations, feature selection, and hyperparameter choices are trained without opening the final evaluation data.

C. Assumptions and hypotheses

The study assumes that at least one eligible source can document phenotype measurement provenance, calendar period, site, and the core baseline variables. It does not assume that all sites measure the same trait with the same instrument. Instead, harmonization is an eligibility and modeling requirement. Missing exposure data are not assumed to be random; the protocol requires a missingness description, complete-case comparison, and a preregistered imputation sensitivity analysis.

H1 states that, for at least one pre-specified endpoint, the pathway-aware model will improve a proper score and probability calibration on both a temporal and an external-site holdout. **H2** states that separating congenital agenesis from treatment and post-developmental dental outcomes will materially alter descriptive interpretation compared with an undifferentiated missing-wisdom-tooth label. Neither hypothesis predicts the sign of a cohort trend. Both fail if the registered gates are not met.

VI. STUDY DESIGN AND METHODS

A. Data eligibility, consent, and data minimization

The protocol is noninterventive. It may use only existing longitudinal cohorts or prospectively consented collections with a clear ethics approval, lawful data-use basis, and a specific prohibition on individual appearance prediction. The final source list, participant count, age range, follow-up duration, and phenotype coverage are human-review inputs. They are not invented in this proposal.

The data custodian must supply a data dictionary that includes endpoint provenance, measurement protocol, calendar period, site, referral or sampling pathway, and missingness indicators. Direct identifiers, raw facial images, and free-text descriptions of appearance are not analytic deliverables. Craniofacial data, when eligible, are reduced to pre-specified identified scalar measurements in an approved secure environment. Genotype data, if considered, require separate consent and are never exported as a score tied to an identifiable person.

B. Phenotype adjudication and measurement harmonization

An endpoint-adjudication manual is completed before data analysis. For the dental case study, a clinician with relevant expertise independently reviews the coding rules and a random, preregistered sample of eligible source records within the secure setting. Disagreements are recorded, not silently resolved. The protocol does not ask for radiation or imaging beyond the recorded care pathway.

For anthropometry, the manual specifies instrument make, calibration procedure, posture, time of day where available, units, and allowable transformation. For craniofacial scalars,

it specifies landmarks, measurement method, version, and a tolerance for inter-rater or algorithm-versus-human agreement. If an instrument change produces an unresolvable discontinuity, the affected records are separated, modeled as a different measurement era, or excluded from confirmatory analysis. A larger dataset is not worth an invalid target.

C. Model development and leakage prevention

For each endpoint, the baseline (2) is first fit using training records. The pathway-aware model (3) is then fit with exactly the pre-specified blocks available before the prediction horizon. Any preprocessing operation that learns from a distribution, including normalization, imputation, missingness-indicator construction, and feature selection, is fit within the training fold and applied to held-out data. Site and family identifiers are used to prevent the same participant or close family group, where known, from appearing on both sides of an evaluation split. This avoids artificial performance created by record overlap.

The model class is chosen according to endpoint type. A binary dental endpoint can use a hierarchical logistic model or an equally transparent pre-registered alternative. A continuous scalar endpoint can use a hierarchical continuous-outcome model with residual and interval calibration. Complex machine-learning alternatives are permitted only when they use the same target, time horizon, predictor availability, and splits as the baseline. They cannot obtain an apparent gain by reading variables recorded after the outcome or by exploiting a source-specific identifier.

The optional genetic block is evaluated only after a non-genetic pathway-aware model has been analyzed. Its contribution is reported as an incremental comparison and must be accompanied by sensitivity to ancestry structure, relatedness, covariate specification, and site composition. A failure of this block does not invalidate the broader study; it demonstrates that the block should not be used. This conforms to the reason for including the block as optional in the first place: a potentially informative source of variation is not automatically a transportable or interpretable predictor.

D. Validation, calibration, and uncertainty

The primary evaluation is a locked temporal holdout, such as a later measurement era or later birth cohort selected before model fitting. The secondary evaluation is an external-site holdout. Both are essential because a model that describes one historical source may fail when an instrument, care setting, population composition, or exposure distribution changes. Separate within-site random splitting is insufficient for the scientific question because it can preserve the same measurement quirks on both sides of the split.

For a binary endpoint with outcome A_i and predicted probability $\hat{p}_i$, one registered proper score is the Brier score:

$$BS = \frac{1}{n} \sum_{i=1}^n (A_i - \hat{p}_i)^2. \quad (5)$$

TABLE II
PRE-ANALYSIS REGISTER FOR PHENOTYPE FAMILIES AND PATHWAY BLOCKS

Element	Required operational record	Primary purpose	Exclusion or downgrade trigger
Third-molar agenesis	Age-adequate developmental evidence; extraction, eruption, impaction, and treatment fields	Bounded congenital dental outcome	Absence cannot be separated from post-developmental tooth status.
Craniofacial scalar family	Approved de-identified measurement definitions; landmark and protocol version	Quantitative morphology without releasing faces	Raw images are needed, landmark method changes without bridge data, or agreement falls below the registered tolerance.
Anthropometric family	Instrument, units, protocol, and measurement era	Test a standard bodily measure with explicit process adjustment	Unit or protocol cannot be harmonized or modeled.
Developmental block	Time-aligned, consented exposure data with provenance and missingness fields	Separate ontogenetic context from cohort labels	Timing or consent is insufficient.
Technology and treatment block	Dental and health treatment history or documented access proxy	Prevent treatment from masquerading as congenital biology	Variable cannot distinguish pre- and post-outcome timing.
Optional inherited block	Separate consent, genotype QC, relatedness and structure records	Sensitivity-tested incremental prediction only	Structure, transferability, or subgroup calibration analysis is inadequate.

In (5), n is the size of the locked evaluation set. Lower scores are better, but a difference is interpretable only with an uncertainty interval and comparison against the baseline. Calibration intercept and slope are reported with the reliability display. For continuous endpoints, the protocol uses an outcome-appropriate proper score, residual calibration, and predictive-interval coverage. A ranking metric alone is not accepted because good ranking can coexist with systematically overconfident predictions.

TRIPOD motivates complete disclosure of model development and validation [6]; PROCAST supplies a structured audit of participants, predictors, outcomes, and analysis [7]. In this protocol, these standards are not treated as badges. Their logic is embedded in the locked target, the data dictionary, the split plan, and the release gates.

E. Sample support, missingness, and sensitivity planning

No numerical sample size is fabricated before the eligible cohorts are known. Instead, the statistical review pre-registers an outcome-specific simulation plan once the prevalence, number of sites, correlation structure, and allowed degrees of freedom are documented. The plan estimates the precision of calibration and proper-score differences under the actual candidate data geometry. If an endpoint or a prespecified stratum lacks support for a meaningful calibration assessment, the result is labeled insufficiently supported rather than reported with a falsely precise estimate.

Missing data are potentially informative. An absent treatment history may mean that no treatment occurred, that care was delivered outside the record, or that the field was not collected during that era. The analysis records which of these mechanisms is plausible for every variable. It compares a complete-case analysis with a preregistered imputation approach and with missingness indicators where justified. A missingness procedure is not allowed to use held-out outcomes. Material disagreement among the analyses is reported as a limitation, not hidden by selecting the most favorable result.

The robustness package includes leave-one-site-out evaluation, leave-one-measurement-era-out evaluation, removal of each pathway block, and negative controls when a defensible control can be defined. For example, a purported developmental-window exposure should not show the same association with a temporally impossible proxy. A negative control does not identify the causal effect of an exposure; it can reveal that an association is compatible with measurement or composition artifact.

F. Reproducibility and preregistration

Before opening the final temporal or site holdout, the study archives a versioned analysis plan, data dictionary, endpoint adjudication manual, feature availability table, split definitions, performance metrics, and interpretation rules. Code is executed only in the approved analysis environment. It produces a lineage record for input versions, transformations, model parameters, and tables. No raw facial image, person-level genotype, or identifiable record is included in a public replication package. A synthetic or metadata-only demonstration may be released only after governance review confirms that it cannot be mistaken for observed study results.

VII. EXPECTED OUTCOME BRANCHES AND CONDITIONAL CONCLUSIONS

The study is planned so that several outcomes are informative. The first branch is *no incremental predictive value*. If the pathway-aware model does not improve temporal and external-site calibration over baseline, the result shows that the available records do not support an added pathway-aware forecast. It does not show that pathways are biologically irrelevant. The limitation may be poor measurement, inappropriate time alignment, limited exposure range, or an endpoint that is not transportable.

The second branch is *baseline explains apparent cohort trend*. Suppose an unadjusted cohort association becomes weak or disappears after measurement era and demographic-composition adjustment. Then the appropriate conclusion is

TABLE III
RELEASE GATES FOR ANY CONDITIONAL FORECAST STATEMENT

Gate	Requirement
Endpoint validity	Confirmatory endpoint meets its pre-analysis definition and provenance rules.
Temporal transport	Pathway-aware model meets registered performance and calibration criteria in the locked later-period split.
Site transport	The result persists on an independent-site hold-out or is explicitly downgraded as site-specific.
Robustness	Measurement-era, composition, missing-data, and block-ablation analyses do not negate the claim.
Genetic caution	Optional inherited block passes structure, relatedness, transferability, and subgroup-calibration review; otherwise it is removed.
Governance	Consent, privacy, specialist, statistical, and communications reviews approve the wording and release.

that the observed difference was not robust to non-biological competing explanations. It must not be described as evidence against evolution in all contexts; it is a result about this phenotype, dataset, and model.

The third branch is *developmental or technology block improves prediction*. If an exposure or treatment block adds stable, calibrated predictive value, the conclusion may be that the block carries useful population-level information in the eligible setting. The conclusion remains associational. A separate causal design would be needed before claiming that changing an exposure or treatment changes a phenotype.

The fourth branch is *optional genetic block is unstable*. This is an expected and useful possibility. If performance varies strongly with population-structure controls, site, relatedness, or subgroup, the block is not used in a forecast. That outcome implements, rather than merely cites, the caution raised by Berg *et al.* [4].

The fifth branch is *limited, transportable pathway attribution*. If every gate passes for a clearly defined endpoint, the study may report a conditional statement such as: in the eligible populations and measured periods, a specified combination of measured pathway blocks improves calibrated description of the phenotype over the baseline. It still cannot predict an individual’s appearance, assert inevitability across all humans, or imply a hierarchy among human groups.

VIII. RISKS, LIMITATIONS, AND HUMAN REVIEW REQUIREMENTS

A. Scientific and statistical limitations

The central scientific limitation is incomplete identification. Multiple pathways can co-vary: an exposure can differ by site and period; treatment availability can change with social conditions; and measured composition can be an imperfect proxy for unmeasured history. The model comparison is useful for prediction and competing-explanation audit, but it does not solve causal identification. A causal effect of diet, technology, or an environmental exposure would require a distinct target

trial emulation, natural experiment, or other justified design that is outside this proposal.

Measurement remains a decisive limitation. Craniofacial morphology is particularly sensitive to landmark definitions, imaging method, posture, age, and selection pathway. Dental records can be referral-biased and often lack complete extraction history. An apparently large, multisite dataset can still be less informative than a smaller, well-provenanced dataset. The protocol therefore accepts a conservative result or no confirmatory analysis when harmonization cannot be demonstrated.

Genetic interpretation is also constrained. An optional inherited-feature summary may be less portable across cohorts, may encode residual population structure, and may fail to improve calibration. It is not a measure of human worth, biological identity, or a personal destiny. No individual-level result is returned, and no model is deployed for clinical, educational, employment, insurance, reproductive, policing, or immigration decisions.

B. Ethics, privacy, and communication

Human physical traits and genomic information are sensitive. The study requires research-ethics approval, a lawful data-use basis, data minimization, and a reidentification assessment. Facial images are excluded from released analysis artifacts. Any linkage of phenotype and genotype uses approved, pseudonymized records in a controlled environment. Access logging, retention limits, and a process for withdrawal or custodial restrictions must be agreed with the data holder before analysis.

Communication is not a cosmetic final step. A result that is technically well calibrated can still be misrepresented as a story about racial destiny or a universal future body. The release gate therefore includes review by domain specialists, quantitative-methods reviewers, data-governance personnel, and reviewers qualified to identify deterministic or stigmatizing language. The public-facing summary must distinguish association, prediction, causal inference, and inherited evolution.

C. Scope boundary

This proposal does not test whether humans will look different in a singular, inevitable sense. It provides a way to ask narrower questions responsibly. For each phenotype, the protocol can distinguish evidence for a measured population-level association from a claim about inherited evolution, and it can identify when the available evidence is too weak to support a forecast. That is a strength, not an evasion: uncertainty is part of the scientific result.

IX. CONCLUSION

Future human physical phenotype cannot be responsibly inferred from a single narrative about evolution, diet, climate, technology, or wisdom teeth. Different generative pathways can yield similar visible patterns while implying different scientific and ethical conclusions. The Phenotype Pathway

TABLE IV
HUMAN REVIEW RESPONSIBILITIES AND STOP CONDITIONS

Review role	Required decision	Stop or downgrade condition
Research ethics and data governance	Confirm lawful basis, consent scope, minimization, retention, and reidentification controls.	Consent or data-use basis is unclear; raw images or identifiable genetic outputs are required.
Dental and craniofacial specialists	Approve endpoint manual, age adequacy, and measurement harmonization.	Agensis cannot be separated from extraction or treatment; landmark definitions are not comparable.
Population-genetic specialist	Review genotype QC, structure, relatedness, transferability, and interpretation.	Optional inherited block remains structurally confounded or harms calibration in a supported stratum.
Statistician and reproducibility reviewer	Approve split logic, missing-data plan, metrics, code review, and preregistration.	Holdout leakage, unregistered target change, or unresolvable performance instability.
Communications and equity reviewer	Approve language and suppress disallowed inferences.	Wording invites individual prediction, population ranking, biological determinism, or eugenic interpretation.

Attribution Atlas converts that recognition into a testable research design. It defines bounded endpoints, separates congenital dental status from treatment history, uses a cohort-and-measurement baseline, adds pathway blocks conservatively, and requires temporal and external-site calibration before any conditional forecast.

The proposal is intentionally strict about genetics and communication. Inherited summaries are optional, consented, and sensitivity-tested; they are never a population ranking or an individual appearance device. Technology and developmental exposures are interpreted as associations unless a separate causal design is justified. If the study fails its validation or governance gates, it reports that failure without inventing a future-human story. If it succeeds, its conclusion remains local to the measured phenotype, eligible populations, periods, and modeling assumptions. This is the appropriate scientific answer to a broad question: physical differences may occur, but their pathways must be measured rather than imagined.

APPENDIX A IDEA-EVOLUTION RECORD

The workflow required a Survey–Idea–ExperimentDesign–Author provenance chain. The Survey froze evidence and three gaps before idea selection. IDEA-FP-01, a secular trend catalogue, was retained as a descriptive alternative but rejected as the primary direction because it cannot separate the relevant pathways. IDEA-FP-02, the Phenotype Pathway Attribution Atlas, was selected because it ties each accepted gap to a measurable variable block and a falsification condition. IDEA-FP-03, a universal future-human visual generator, was rejected because it has no defensible universal target and would create deterministic and stereotype risks.

No autonomous search trajectory, MCTS trace, experiment, simulation, or multi-agent debate was executed. This appendix records the actual route selection rather than fabricating an optimization history. The selected idea remains a proposal: its central hypothesis is not that it has already improved any metric, but that its benefit can be tested using locked temporal and external-site holdouts.

TABLE V
SYMBOLS AND DEFINITIONS

Symbol or term	Definition
Y_{ijk}	Phenotype k for person i at visit or age interval j .
A_i	Confirmatory binary indicator of documented congenital third-molar agensis, defined in (1).
$\eta_{ijk}^{(0)}$	Baseline model linear predictor containing age or stage, cohort, site, and measurement process.
$\eta_{ijk}^{(1)}$	Pathway-aware model linear predictor in (3).
$\mathbf{x}_{ij}$	Demographic-composition predictor block.
$\mathbf{z}_{ij}$	Time-aligned developmental-exposure predictor block.
$\mathbf{q}_{ij}$	Technology or treatment predictor block.
$\mathbf{r}_i$	Optional separately consented inherited-feature summary.
Calibration	Agreement between predicted probabilities or intervals and observed frequencies or outcomes in held-out data.
Transportability	Preservation of registered predictive performance across later periods or independent sites.

APPENDIX B SYMBOLS AND OPERATIONAL DEFINITIONS

APPENDIX C EVIDENCE COVERAGE AND REVIEW CHECKLIST

Table VI shows the allowed evidence role for each frozen source. Literature sources motivate only the claims shown in the table; they do not provide observed results for this study. The Author stage adds no citations outside the Survey registry and no acknowledgements, personal contributions, simulations, or result claims.

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TABLE VI
EVIDENCE COVERAGE AND AUTHOR-STAGE CONSTRAINT CHECK

Source	Frozen evidence role	Permitted Author-stage use
Ponzi <i>et al.</i> [1]	Developmental plasticity during ontogeny.	Separate developmental context from inherited-change claims.
Carter <i>et al.</i> [2]	Heterogeneous prevalence of third-molar agenesis.	Justify a bounded dental endpoint and prohibit trend or selection overclaim.
Laland <i>et al.</i> [3]	Gene–culture coevolution framework.	Model culture and technology as conditional environmental pathways.
Berg <i>et al.</i> [4]	Population-stratification risk in polygenic adaptation inference.	Require conservative genetic sensitivity analysis.
1000 Genomes Project [5]	Human genetic variation resource.	Require QC and documented structure for any optional genetic block.
Collins <i>et al.</i> [6]	Transparent prediction-model reporting.	Lock target, predictors, validation, and reporting.
Wolff <i>et al.</i> [7]	Prediction-model bias and applicability domains.	Audit participants, predictors, outcome, and analysis.

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